Predicting Insurance Cost

Insurance Cost Analysis

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```r
#dataset link: https://www.kaggle.com/datasets/mirichoi0218/insurance
# it was further appended using AI
library(MASS)
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

```r
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.4.3

##
## Attaching package: 'dplyr'

## The following object is masked from 'package:MASS':
##
##     select

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(tidyr)
```

```
## Warning: package 'tidyr' was built under R version 4.4.3
```

```r
library(GGally)
library(car)
```

```
## Warning: package 'car' was built under R version 4.4.3

## Loading required package: carData
```

```
## Warning: package 'carData' was built under R version 4.4.3

##
## Attaching package: 'car'

## The following object is masked from 'package:dplyr':
##
##     recode
```

Load the Dataset

```r
# "insurance_extended.csv"
# "insurance_realistic_augmented.csv"
# "insurance.csv"
df <- read.csv("insurance.csv")
```

Data Cleaning

```r
sum(is.na(df))
```

```
## [1] 0
```

```r
summary(df)
```

```
##       age            sex                 bmi           children
##  Min.   :18.00   Length:1338        Min.   :15.96   Min.   :0.000
##  1st Qu.:27.00   Class :character   1st Qu.:26.30   1st Qu.:0.000
##  Median :39.00   Mode  :character   Median :30.40   Median :1.000
##  Mean   :39.21                      Mean   :30.66   Mean   :1.095
##  3rd Qu.:51.00                      3rd Qu.:34.69   3rd Qu.:2.000
##  Max.   :64.00                      Max.   :53.13   Max.   :5.000
##     smoker             region             charges
##  Length:1338        Length:1338        Min.   : 1122
##  Class :character   Class :character   1st Qu.: 4740
##  Mode  :character   Mode  :character   Median : 9382
##                                        Mean   :13270
##                                        3rd Qu.:16640
##                                        Max.   :63770
```

```r
# Nothing looks off
```

Exploratory Data Analysis

```r
ggpairs(
  df |> select(
   age,
   bmi,
   charges
  ),
  columns = 1:3,
  aes(alpha = 0.6),
  upper = list(
    continuous = wrap(
      "cor",
      digits = 2,
      size = 2.2,
      mapping = aes()
    )
  ),
  lower = list(
    continuous = wrap("points", size = 0.5, color = "coral")
  ),
  diag = list(
    continuous = wrap("barDiag", bins = 20, fill = "coral")
  ),
  progress = FALSE
)+ labs(caption = "Figure 2: Scatterplot matrix bivariate and univariate distributions") +
   theme(
    plot.caption = element_text(hjust = 0)
  )
```

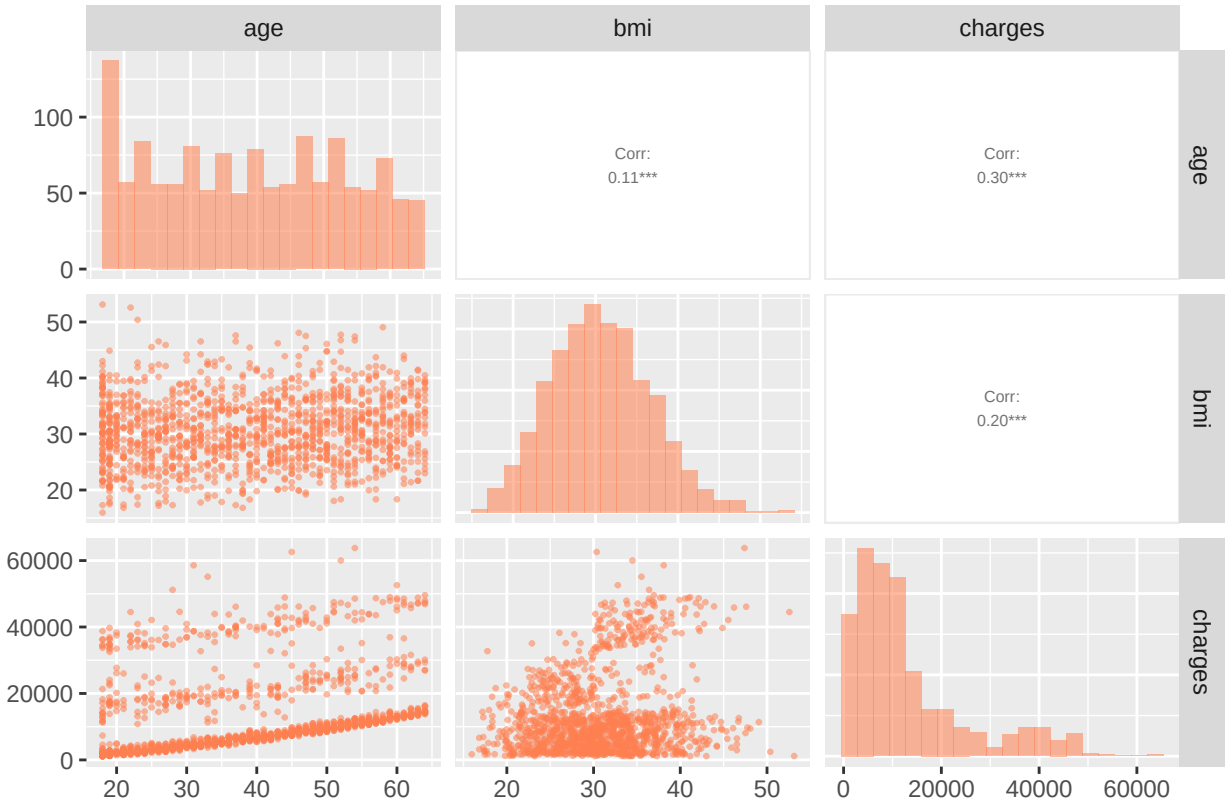
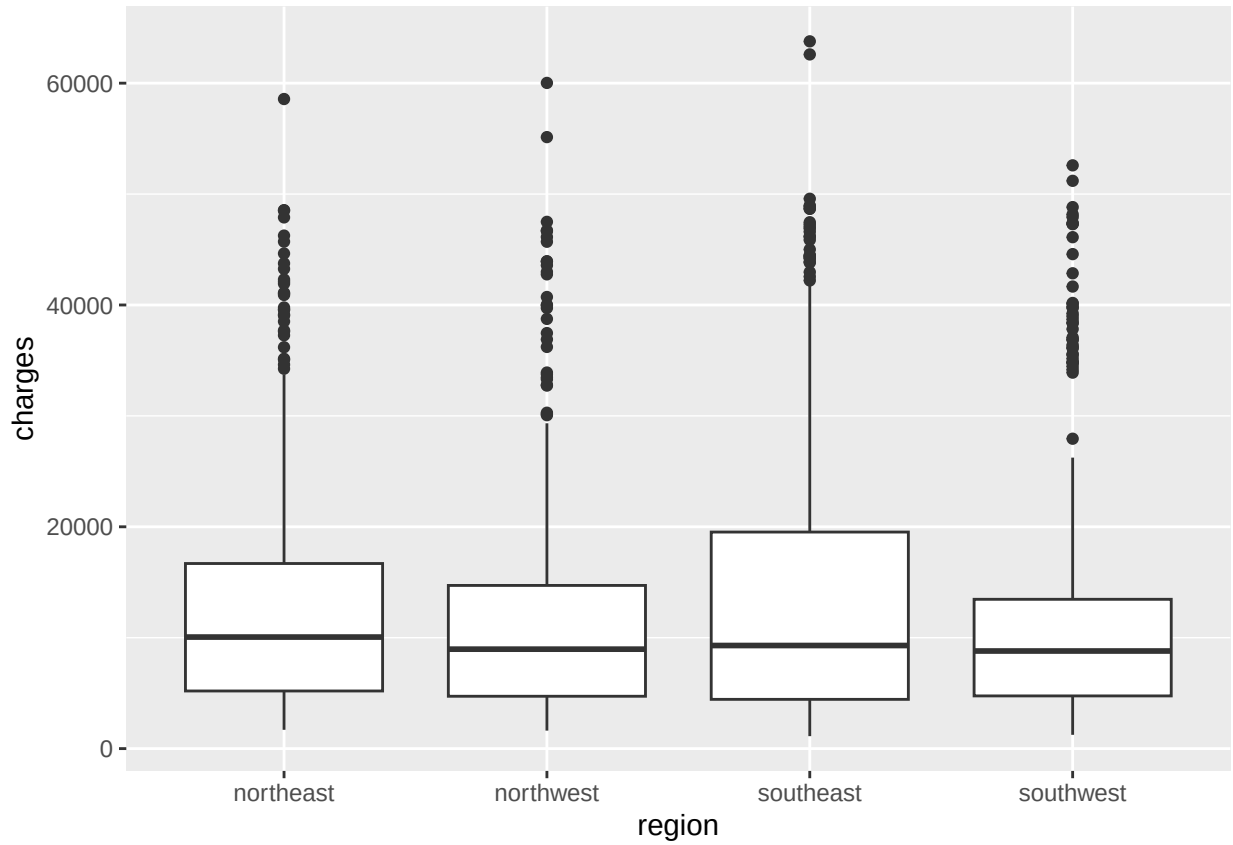


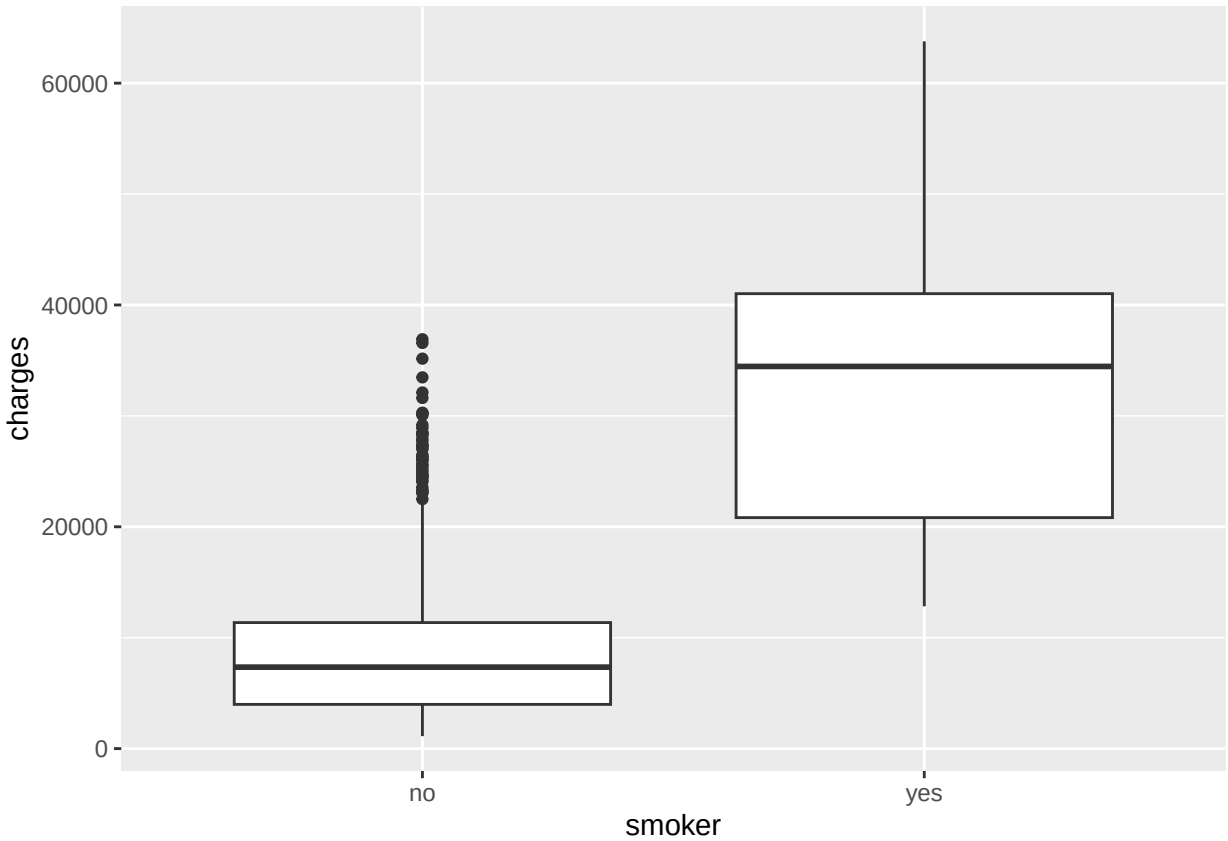
Figure 2: Scatterplot matrix bivariate and univariate distributions

Charges and Age have a positive linear correlation. There is also a roughly positive linear relationship between BMI and charges, as well as Age and BMI. Age is roughly reminiscent of a uniform distribution, while the distribution closest to normal is BMI. Weak correlation observed between the predictors, which ensures that each predictor brings new information to the model.

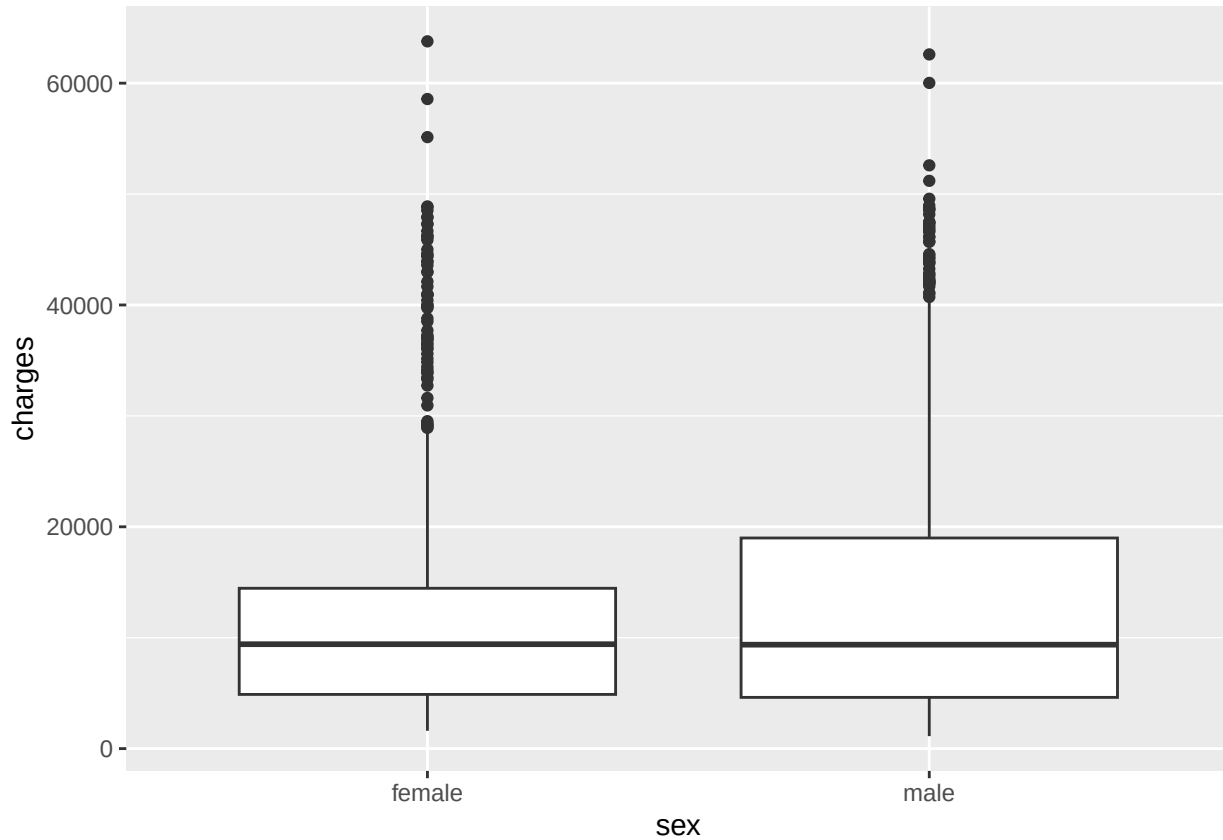
```
df |>
  ggplot(aes(x = region, y= charges)) +
  geom_boxplot()
```




```
df |>
  ggplot(aes(x = smoker, y= charges)) +
  geom_boxplot()
```




```
df |>
  ggplot(aes(x = sex, y= charges)) +
  geom_boxplot()
```



The biggest discrepancy of mean insurance cost can be observed between smoking and non-smoking groups.

Multiple Linear Regression Model

```
# split data into train and test

set.seed(123) # for reproducibility
train_index <- sample(1:nrow(df), 0.8 * nrow(df))
train_data <- df[train_index, ]
test_data <- df[-train_index, ]


# validating the split
nrow(train_data) / nrow(df)
```

```
## [1] 0.799701
```

```
nrow(test_data) / nrow(df)
```

```
## [1] 0.200299
```

```r
# run regression model
model <- lm(charges ~ (.)^2, data = train_data)
# also (.)^2 includes main effects AND all 2-way interactions

# used to be model <- lm(charges ~ (.)^2, data = df) but i changed it becuase there's no point in split

# after getting our (.)^2 model, we perform stepwise model selection, picking a model based on the AIC.

summary(model)
```

```
##
## Call:
## lm(formula = charges ~ (.)^2, data = train_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12065.9  -2011.2  -1205.1   -157.8  29967.1
##
## Coefficients:
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                -2.611e+03  3.033e+03  -0.861  0.38962
## age                         2.077e+02  6.194e+01   3.352  0.00083 ***
## sexmale                     2.436e+02  1.829e+03   0.133  0.89407
## bmi                         8.132e+01  9.835e+01   0.827  0.40855
## children                    1.407e+03  7.958e+02   1.768  0.07737 .
## smokeryes                  -2.034e+04  2.200e+03  -9.244  < 2e-16 ***
## regionnorthwest            -2.383e+03  2.628e+03  -0.907  0.36460
## regionsoutheast             2.374e+03  2.483e+03   0.956  0.33937
## regionsouthwest            -1.620e+03  2.546e+03  -0.636  0.52462
## age:sexmale                 5.993e+00  2.220e+01   0.270  0.78726
## age:bmi                     4.799e-01  1.903e+00   0.252  0.80100
## age:children               -3.034e+00  1.008e+01  -0.301  0.76354
## age:smokeryes              -3.035e+00  2.725e+01  -0.111  0.91134
## age:regionnorthwest         2.298e+01  3.174e+01   0.724  0.46927
## age:regionsoutheast         5.060e+01  3.167e+01   1.598  0.11043
## age:regionsouthwest         5.820e+01  3.244e+01   1.794  0.07314 .
## sexmale:bmi                -1.855e+01  5.315e+01  -0.349  0.72718
## sexmale:children           -3.525e+02  2.560e+02  -1.377  0.16884
## sexmale:smokeryes          -3.201e+02  7.655e+02  -0.418  0.67591
## sexmale:regionnorthwest    -5.861e+02  8.725e+02  -0.672  0.50190
## sexmale:regionsoutheast     4.932e+02  8.765e+02   0.563  0.57378
## sexmale:regionsouthwest     8.799e+01  8.809e+02   0.100  0.92046
## bmi:children               -1.731e+01  2.240e+01  -0.773  0.43978
## bmi:smokeryes               1.466e+03  6.303e+01  23.257  < 2e-16 ***
## bmi:regionnorthwest         2.793e+01  7.985e+01   0.350  0.72654
## bmi:regionsoutheast        -1.722e+02  6.949e+01  -2.478  0.01337 *
## bmi:regionsouthwest        -5.900e+01  7.747e+01  -0.762  0.44651
## children:smokeryes         -1.597e+02  3.303e+02  -0.483  0.62895
## children:regionnorthwest    2.815e+02  3.661e+02   0.769  0.44217
## children:regionsoutheast   -1.025e+01  3.628e+02  -0.028  0.97746
## children:regionsouthwest   -3.996e+02  3.535e+02  -1.130  0.25854
## smokeryes:regionnorthwest -7.825e+02  1.073e+03  -0.729  0.46610
## smokeryes:regionsoutheast -8.109e+02  1.049e+03  -0.773  0.43948
```

```
## smokeryes:regionsouthwest  8.847e+02  1.099e+03   0.805  0.42099
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4913 on 1036 degrees of freedom
## Multiple R-squared:  0.846,  Adjusted R-squared:  0.8411
## F-statistic: 172.5 on 33 and 1036 DF,  p-value: < 2.2e-16
```

```r
aic_model <- stepAIC(model, direction = "both")
```

```
## Start:  AIC=18222.65
## charges ~ (age + sex + bmi + children + smoker + region)^2
##
##                   Df  Sum of Sq        RSS   AIC
## - sex:region       3 3.7105e+07 2.5043e+10 18218
## - smoker:region    3 7.7915e+07 2.5084e+10 18220
## - children:region  3 8.8994e+07 2.5095e+10 18220
## - age:smoker       1 2.9940e+05 2.5006e+10 18221
## - age:bmi          1 1.5342e+06 2.5007e+10 18221
## - age:sex          1 1.7587e+06 2.5008e+10 18221
## - age:children     1 2.1856e+06 2.5008e+10 18221
## - sex:bmi          1 2.9394e+06 2.5009e+10 18221
## - sex:smoker       1 4.2208e+06 2.5010e+10 18221
## - age:region       3 9.8260e+07 2.5104e+10 18221
## - children:smoker  1 5.6392e+06 2.5012e+10 18221
## - bmi:children     1 1.4417e+07 2.5020e+10 18221
## - sex:children     1 4.5761e+07 2.5052e+10 18223
## <none>                          2.5006e+10 18223
## - bmi:region       3 2.2698e+08 2.5233e+10 18226
## - bmi:smoker       1 1.3055e+10 3.8061e+10 18670
##
## Step:  AIC=18218.24
## charges ~ age + sex + bmi + children + smoker + region + age:sex +
##     age:bmi + age:children + age:smoker + age:region + sex:bmi +
##     sex:children + sex:smoker + bmi:children + bmi:smoker + bmi:region +
##     children:smoker + children:region + smoker:region
##
##                   Df  Sum of Sq        RSS   AIC
## - smoker:region    3 7.6884e+07 2.5120e+10 18216
## - children:region  3 8.6390e+07 2.5129e+10 18216
## - sex:bmi          1 7.5361e+04 2.5043e+10 18216
## - age:smoker       1 1.2125e+05 2.5043e+10 18216
## - age:bmi          1 1.0566e+06 2.5044e+10 18216
## - age:sex          1 1.5272e+06 2.5045e+10 18216
## - age:children     1 2.4240e+06 2.5045e+10 18216
## - sex:smoker       1 3.2910e+06 2.5046e+10 18216
## - age:region       3 9.8318e+07 2.5141e+10 18216
## - children:smoker  1 7.4642e+06 2.5050e+10 18217
## - bmi:children     1 1.4605e+07 2.5058e+10 18217
## <none>                          2.5043e+10 18218
## - sex:children     1 4.7868e+07 2.5091e+10 18218
## - bmi:region       3 2.2112e+08 2.5264e+10 18222
## + sex:region       3 3.7105e+07 2.5006e+10 18223
## - bmi:smoker       1 1.3037e+10 3.8080e+10 18665
```

```
##
## Step:  AIC=18215.52
## charges ~ age + sex + bmi + children + smoker + region + age:sex +
##     age:bmi + age:children + age:smoker + age:region + sex:bmi +
##     sex:children + sex:smoker + bmi:children + bmi:smoker + bmi:region +
##     children:smoker + children:region
##
##                    Df  Sum of Sq        RSS   AIC
## - children:region  3 8.0081e+07 2.5200e+10 18213
## - age:region       3 8.9143e+07 2.5209e+10 18213
## - sex:bmi          1 1.5483e+05 2.5120e+10 18214
## - age:smoker       1 8.3793e+05 2.5121e+10 18214
## - age:sex          1 8.6980e+05 2.5121e+10 18214
## - age:bmi          1 9.0938e+05 2.5121e+10 18214
## - sex:smoker       1 1.2676e+06 2.5121e+10 18214
## - age:children     1 1.2942e+06 2.5121e+10 18214
## - children:smoker  1 5.2728e+06 2.5125e+10 18214
## - bmi:children     1 1.7627e+07 2.5138e+10 18214
## <none>                          2.5120e+10 18216
## - sex:children     1 5.2548e+07 2.5172e+10 18216
## + smoker:region    3 7.6884e+07 2.5043e+10 18218
## - bmi:region       3 2.2158e+08 2.5341e+10 18219
## + sex:region       3 3.6074e+07 2.5084e+10 18220
## - bmi:smoker       1 1.4340e+10 3.9460e+10 18697
##
## Step:  AIC=18212.92
## charges ~ age + sex + bmi + children + smoker + region + age:sex +
##     age:bmi + age:children + age:smoker + age:region + sex:bmi +
##     sex:children + sex:smoker + bmi:children + bmi:smoker + bmi:region +
##     children:smoker
##
##                    Df  Sum of Sq        RSS   AIC
## - age:region       3 8.7845e+07 2.5288e+10 18211
## - sex:bmi          1 4.8320e+04 2.5200e+10 18211
## - age:children     1 5.9177e+05 2.5201e+10 18211
## - sex:smoker       1 7.0276e+05 2.5201e+10 18211
## - age:sex          1 1.1386e+06 2.5201e+10 18211
## - age:smoker       1 1.3741e+06 2.5201e+10 18211
## - age:bmi          1 1.8832e+06 2.5202e+10 18211
## - children:smoker  1 4.5426e+06 2.5204e+10 18211
## - bmi:children     1 1.9659e+07 2.5220e+10 18212
## <none>                          2.5200e+10 18213
## - sex:children     1 5.4290e+07 2.5254e+10 18213
## + children:region  3 8.0081e+07 2.5120e+10 18216
## + smoker:region    3 7.0575e+07 2.5129e+10 18216
## - bmi:region       3 2.2726e+08 2.5427e+10 18216
## + sex:region       3 3.3970e+07 2.5166e+10 18218
## - bmi:smoker       1 1.4412e+10 3.9612e+10 18695
##
## Step:  AIC=18210.65
## charges ~ age + sex + bmi + children + smoker + region + age:sex +
##     age:bmi + age:children + age:smoker + sex:bmi + sex:children +
##     sex:smoker + bmi:children + bmi:smoker + bmi:region + children:smoker
##
```

```
##                    Df  Sum of Sq        RSS   AIC
## - sex:bmi           1 1.4064e+05 2.5288e+10 18209
## - age:children      1 3.2888e+05 2.5288e+10 18209
## - sex:smoker        1 5.7893e+05 2.5288e+10 18209
## - age:smoker        1 8.1680e+05 2.5289e+10 18209
## - age:sex           1 9.5508e+05 2.5289e+10 18209
## - children:smoker   1 6.2949e+06 2.5294e+10 18209
## - age:bmi           1 1.1514e+07 2.5299e+10 18209
## - bmi:children      1 2.0661e+07 2.5308e+10 18210
## <none>                           2.5288e+10 18211
## - sex:children      1 4.8993e+07 2.5337e+10 18211
## + age:region        3 8.7845e+07 2.5200e+10 18213
## + children:region   3 7.8784e+07 2.5209e+10 18213
## - bmi:region        3 2.1140e+08 2.5499e+10 18214
## + smoker:region     3 6.2241e+07 2.5226e+10 18214
## + sex:region        3 3.3936e+07 2.5254e+10 18215
## - bmi:smoker        1 1.4351e+10 3.9638e+10 18690
##
## Step:  AIC=18208.65
## charges ~ age + sex + bmi + children + smoker + region + age:sex +
##     age:bmi + age:children + age:smoker + sex:children + sex:smoker +
##     bmi:children + bmi:smoker + bmi:region + children:smoker
##
##                    Df  Sum of Sq        RSS   AIC
## - age:children      1 3.1726e+05 2.5288e+10 18207
## - sex:smoker        1 5.7433e+05 2.5289e+10 18207
## - age:smoker        1 7.8818e+05 2.5289e+10 18207
## - age:sex           1 8.8506e+05 2.5289e+10 18207
## - children:smoker   1 6.4980e+06 2.5294e+10 18207
## - age:bmi           1 1.1725e+07 2.5300e+10 18207
## - bmi:children      1 2.0573e+07 2.5309e+10 18208
## <none>                           2.5288e+10 18209
## - sex:children      1 4.8941e+07 2.5337e+10 18209
## + sex:bmi           1 1.4064e+05 2.5288e+10 18211
## + age:region        3 8.7938e+07 2.5200e+10 18211
## + children:region   3 7.8606e+07 2.5209e+10 18211
## - bmi:region        3 2.1519e+08 2.5503e+10 18212
## + smoker:region     3 6.2292e+07 2.5226e+10 18212
## + sex:region        3 3.1302e+07 2.5257e+10 18213
## - bmi:smoker        1 1.4351e+10 3.9639e+10 18688
##
## Step:  AIC=18206.67
## charges ~ age + sex + bmi + children + smoker + region + age:sex +
##     age:bmi + age:smoker + sex:children + sex:smoker + bmi:children +
##     bmi:smoker + bmi:region + children:smoker
##
##                    Df  Sum of Sq        RSS   AIC
## - sex:smoker        1 5.8698e+05 2.5289e+10 18205
## - age:smoker        1 7.9927e+05 2.5289e+10 18205
## - age:sex           1 8.9572e+05 2.5289e+10 18205
## - children:smoker   1 6.4248e+06 2.5295e+10 18205
## - age:bmi           1 1.1650e+07 2.5300e+10 18205
## - bmi:children      1 2.1542e+07 2.5310e+10 18206
## <none>                           2.5288e+10 18207
```

```
## - sex:children     1 4.8641e+07 2.5337e+10 18207
## + age:children     1 3.1726e+05 2.5288e+10 18209
## + sex:bmi          1 1.2902e+05 2.5288e+10 18209
## + age:region       3 8.7673e+07 2.5201e+10 18209
## + children:region  3 7.8010e+07 2.5210e+10 18209
## - bmi:region       3 2.1487e+08 2.5503e+10 18210
## + smoker:region    3 6.1693e+07 2.5227e+10 18210
## + sex:region       3 3.1439e+07 2.5257e+10 18211
## - bmi:smoker       1 1.4352e+10 3.9641e+10 18686
##
## Step:  AIC=18204.69
## charges ~ age + sex + bmi + children + smoker + region + age:sex +
##     age:bmi + age:smoker + sex:children + bmi:children + bmi:smoker +
##     bmi:region + children:smoker
##
##                    Df  Sum of Sq        RSS   AIC
## - age:smoker       1 7.8398e+05 2.5290e+10 18203
## - age:sex          1 9.3137e+05 2.5290e+10 18203
## - children:smoker  1 6.7726e+06 2.5296e+10 18203
## - age:bmi          1 1.1884e+07 2.5301e+10 18203
## - bmi:children     1 2.2163e+07 2.5311e+10 18204
## <none>                          2.5289e+10 18205
## - sex:children     1 4.8389e+07 2.5337e+10 18205
## + sex:smoker       1 5.8698e+05 2.5288e+10 18207
## + age:children     1 3.2991e+05 2.5289e+10 18207
## + sex:bmi          1 1.2434e+05 2.5289e+10 18207
## + age:region       3 8.7540e+07 2.5201e+10 18207
## + children:region  3 7.7493e+07 2.5211e+10 18207
## - bmi:region       3 2.1430e+08 2.5503e+10 18208
## + smoker:region    3 6.0337e+07 2.5229e+10 18208
## + sex:region       3 3.1194e+07 2.5258e+10 18209
## - bmi:smoker       1 1.4576e+10 3.9865e+10 18690
##
## Step:  AIC=18202.72
## charges ~ age + sex + bmi + children + smoker + region + age:sex +
##     age:bmi + sex:children + bmi:children + bmi:smoker + bmi:region +
##     children:smoker
##
##                    Df  Sum of Sq        RSS   AIC
## - age:sex          1 8.5006e+05 2.5290e+10 18201
## - children:smoker  1 7.1569e+06 2.5297e+10 18201
## - age:bmi          1 1.1779e+07 2.5301e+10 18201
## - bmi:children     1 2.2341e+07 2.5312e+10 18202
## <none>                          2.5290e+10 18203
## - sex:children     1 4.8585e+07 2.5338e+10 18203
## + age:smoker       1 7.8398e+05 2.5289e+10 18205
## + sex:smoker       1 5.7168e+05 2.5289e+10 18205
## + age:children     1 3.4094e+05 2.5289e+10 18205
## + sex:bmi          1 9.7496e+04 2.5290e+10 18205
## + age:region       3 8.6978e+07 2.5203e+10 18205
## + children:region  3 7.7919e+07 2.5212e+10 18205
## - bmi:region       3 2.1385e+08 2.5503e+10 18206
## + smoker:region    3 6.0981e+07 2.5229e+10 18206
## + sex:region       3 3.1065e+07 2.5259e+10 18207
```

```
## - bmi:smoker       1 1.4592e+10 3.9881e+10 18688
##
## Step:  AIC=18200.76
## charges ~ age + sex + bmi + children + smoker + region + age:bmi +
##     sex:children + bmi:children + bmi:smoker + bmi:region + children:smoker
##
##                   Df  Sum of Sq        RSS   AIC
## - children:smoker  1 7.0550e+06 2.5298e+10 18199
## - age:bmi          1 1.2387e+07 2.5303e+10 18199
## - bmi:children     1 2.2791e+07 2.5313e+10 18200
## <none>                          2.5290e+10 18201
## - sex:children     1 4.8107e+07 2.5339e+10 18201
## + age:sex          1 8.5006e+05 2.5290e+10 18203
## + age:smoker       1 7.0266e+05 2.5290e+10 18203
## + sex:smoker       1 6.0610e+05 2.5290e+10 18203
## + age:children     1 3.5150e+05 2.5290e+10 18203
## + sex:bmi          1 4.2918e+04 2.5290e+10 18203
## + age:region       3 8.6789e+07 2.5204e+10 18203
## + children:region  3 7.8126e+07 2.5212e+10 18203
## - bmi:region       3 2.1563e+08 2.5506e+10 18204
## + smoker:region    3 6.0234e+07 2.5230e+10 18204
## + sex:region       3 3.1217e+07 2.5259e+10 18205
## - bmi:smoker       1 1.4600e+10 3.9890e+10 18686
##
## Step:  AIC=18199.06
## charges ~ age + sex + bmi + children + smoker + region + age:bmi +
##     sex:children + bmi:children + bmi:smoker + bmi:region
##
##                   Df  Sum of Sq        RSS   AIC
## - age:bmi          1 1.2075e+07 2.5310e+10 18198
## - bmi:children     1 2.2453e+07 2.5320e+10 18198
## <none>                          2.5298e+10 18199
## - sex:children     1 5.1093e+07 2.5349e+10 18199
## + children:smoker  1 7.0550e+06 2.5290e+10 18201
## + age:smoker       1 1.0711e+06 2.5296e+10 18201
## + sex:smoker       1 9.6118e+05 2.5297e+10 18201
## + age:sex          1 7.4814e+05 2.5297e+10 18201
## + age:children     1 2.7514e+05 2.5297e+10 18201
## + sex:bmi          1 1.8100e+05 2.5297e+10 18201
## + age:region       3 8.8590e+07 2.5209e+10 18201
## + children:region  3 7.7220e+07 2.5220e+10 18202
## - bmi:region       3 2.1722e+08 2.5515e+10 18202
## + smoker:region    3 5.8314e+07 2.5239e+10 18203
## + sex:region       3 3.2494e+07 2.5265e+10 18204
## - bmi:smoker       1 1.4595e+10 3.9893e+10 18684
##
## Step:  AIC=18197.57
## charges ~ age + sex + bmi + children + smoker + region + sex:children +
##     bmi:children + bmi:smoker + bmi:region
##
##                   Df  Sum of Sq        RSS   AIC
## - bmi:children     1 2.1788e+07 2.5331e+10 18196
## <none>                          2.5310e+10 18198
## - sex:children     1 5.3233e+07 2.5363e+10 18198
```

```
## + age:bmi          1 1.2075e+07 2.5298e+10 18199
## + children:smoker  1 6.7424e+06 2.5303e+10 18199
## + age:region       3 9.8693e+07 2.5211e+10 18199
## + age:sex          1 1.3185e+06 2.5308e+10 18200
## + sex:smoker       1 1.2636e+06 2.5308e+10 18200
## + age:smoker       1 9.1052e+05 2.5309e+10 18200
## + sex:bmi          1 3.7492e+05 2.5309e+10 18200
## + age:children     1 2.0951e+05 2.5309e+10 18200
## + children:region  3 7.9792e+07 2.5230e+10 18200
## - bmi:region       3 2.1537e+08 2.5525e+10 18201
## + smoker:region    3 5.7309e+07 2.5252e+10 18201
## + sex:region       3 3.0073e+07 2.5280e+10 18202
## - age              1 1.3103e+10 3.8412e+10 18642
## - bmi:smoker       1 1.4584e+10 3.9893e+10 18682
##
## Step:  AIC=18196.49
## charges ~ age + sex + bmi + children + smoker + region + sex:children +
##     bmi:smoker + bmi:region
##
##                   Df  Sum of Sq        RSS   AIC
## <none>                          2.5331e+10 18196
## - sex:children     1 5.2871e+07 2.5384e+10 18197
## + bmi:children     1 2.1788e+07 2.5310e+10 18198
## + age:bmi          1 1.1409e+07 2.5320e+10 18198
## + children:smoker  1 6.4259e+06 2.5325e+10 18198
## + age:region       3 9.8910e+07 2.5232e+10 18198
## + sex:smoker       1 2.0964e+06 2.5329e+10 18198
## + age:sex          1 1.8329e+06 2.5330e+10 18198
## + age:children     1 1.0933e+06 2.5330e+10 18198
## + age:smoker       1 1.0718e+06 2.5330e+10 18198
## + sex:bmi          1 1.6343e+05 2.5331e+10 18198
## + children:region  3 8.0730e+07 2.5251e+10 18199
## - bmi:region       3 2.0903e+08 2.5540e+10 18199
## + smoker:region    3 5.8432e+07 2.5273e+10 18200
## + sex:region       3 3.0745e+07 2.5301e+10 18201
## - age              1 1.3092e+10 3.8423e+10 18640
## - bmi:smoker       1 1.4600e+10 3.9932e+10 18682
```

```r
summary(aic_model)
```

```
##
## Call:
## lm(formula = charges ~ age + sex + bmi + children + smoker +
##     region + sex:children + bmi:smoker + bmi:region, data = train_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12823.6  -1935.6  -1271.0   -302.6  29850.8
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)          -3184.93    1659.50  -1.919 0.055227 .
## age                    255.07      10.92  23.362  < 2e-16 ***
## sexmale                -70.60     408.71  -0.173 0.862892
```

```
## bmi                     61.11      54.05   1.131 0.258517
## children               704.68     181.78   3.877 0.000112 ***
## smokeryes           -20446.41    1846.34 -11.074  < 2e-16 ***
## regionnorthwest      -2033.77    2343.09  -0.868 0.385599
## regionsoutheast       3816.41    2165.83   1.762 0.078342 .
## regionsouthwest       -370.90    2317.12  -0.160 0.872858
## sexmale:children      -371.87     250.48  -1.485 0.137945
## bmi:smokeryes         1446.77      58.64  24.671  < 2e-16 ***
## bmi:regionnorthwest     42.60      78.66   0.542 0.588225
## bmi:regionsoutheast   -152.49      68.17  -2.237 0.025508 *
## bmi:regionsouthwest    -32.66      75.36  -0.433 0.664789
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4898 on 1056 degrees of freedom
## Multiple R-squared:  0.844,  Adjusted R-squared:  0.8421
## F-statistic: 439.5 on 13 and 1056 DF,  p-value: < 2.2e-16
```

```r
# validate with test set


# 1. Get the actual values from your test set
actuals <- test_data$charges  # Replace with your actual column name

# 2. Get your predictions from your model (you've already done this)
predictions <- predict(aic_model, newdata = test_data)

# 3. Calculate Metrics
mse <- mean((actuals - predictions)^2) # mean squared error
rmse <- sqrt(mse)
mae <- mean(abs(actuals - predictions))
rss <- sum((actuals - predictions)^2)
tss <- sum((actuals - mean(actuals))^2)
r_squared <- 1 - (rss / tss) # r squared



# Calculate MAPE
mape <- mean(abs((actuals - predictions) / actuals)) * 100
cat("The average error is", mape, "%\n")
```

```
## The average error is 27.73041 %
```

```r
cat("The 'Accuracy' is", 100 - mape, "%\n")
```

```
## The 'Accuracy' is 72.26959 %
```

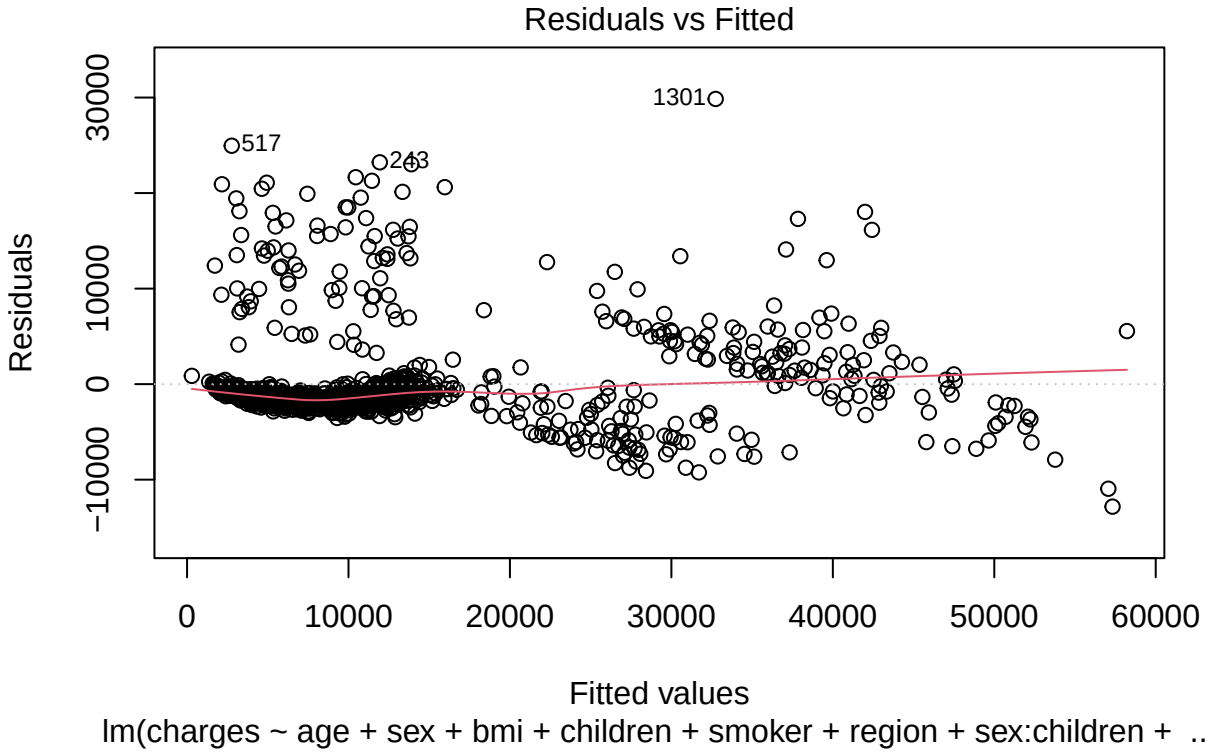
```r
r_squared
```

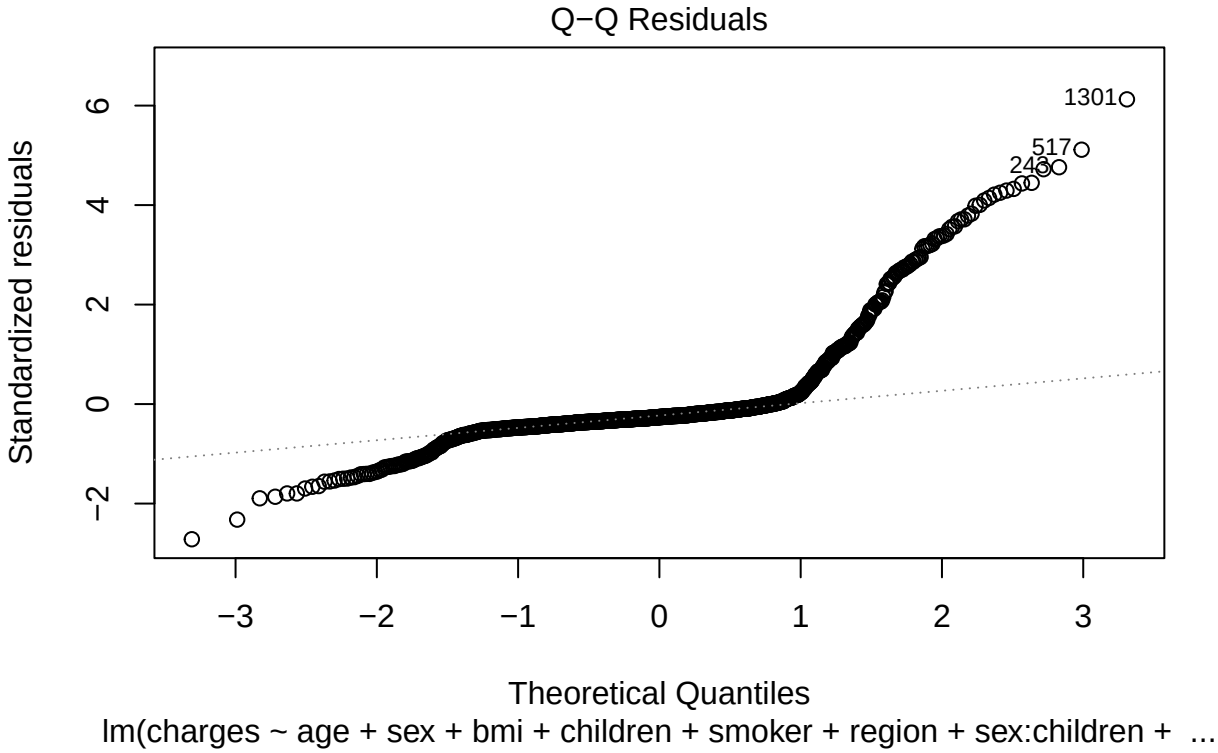
```
## [1] 0.8301909
```

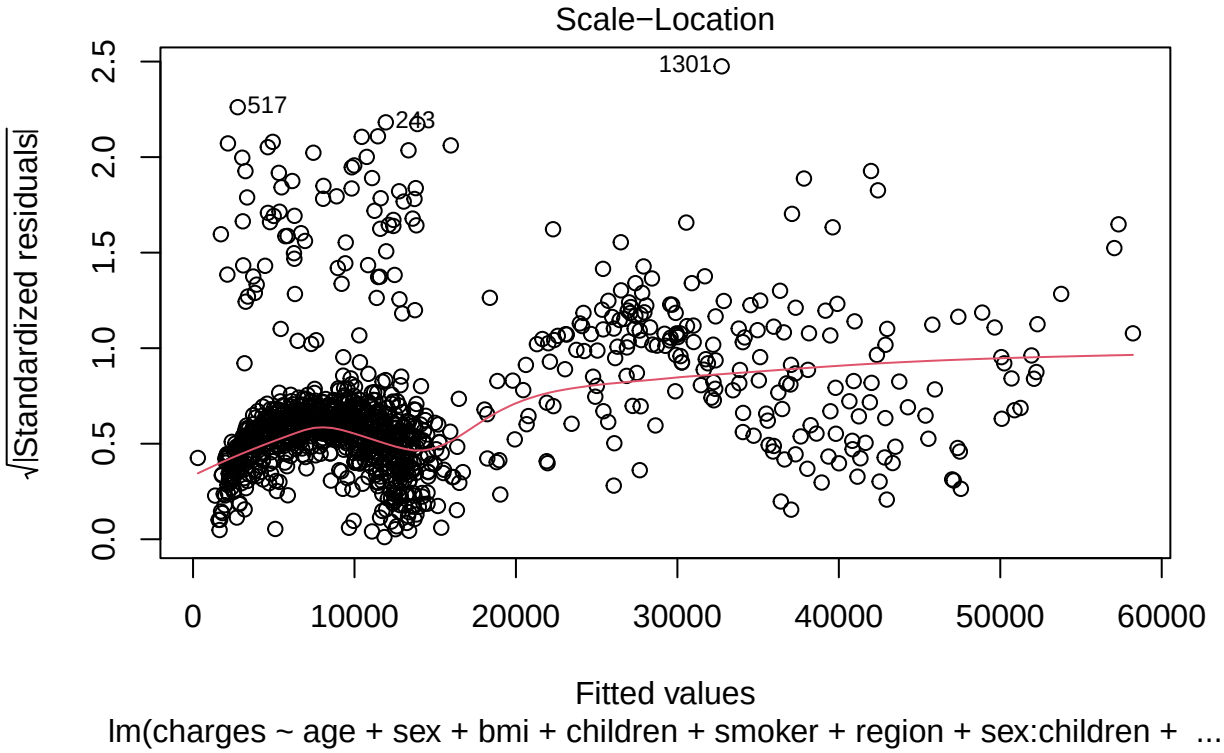
```r
#further model analysis

summary(aic_model)
```

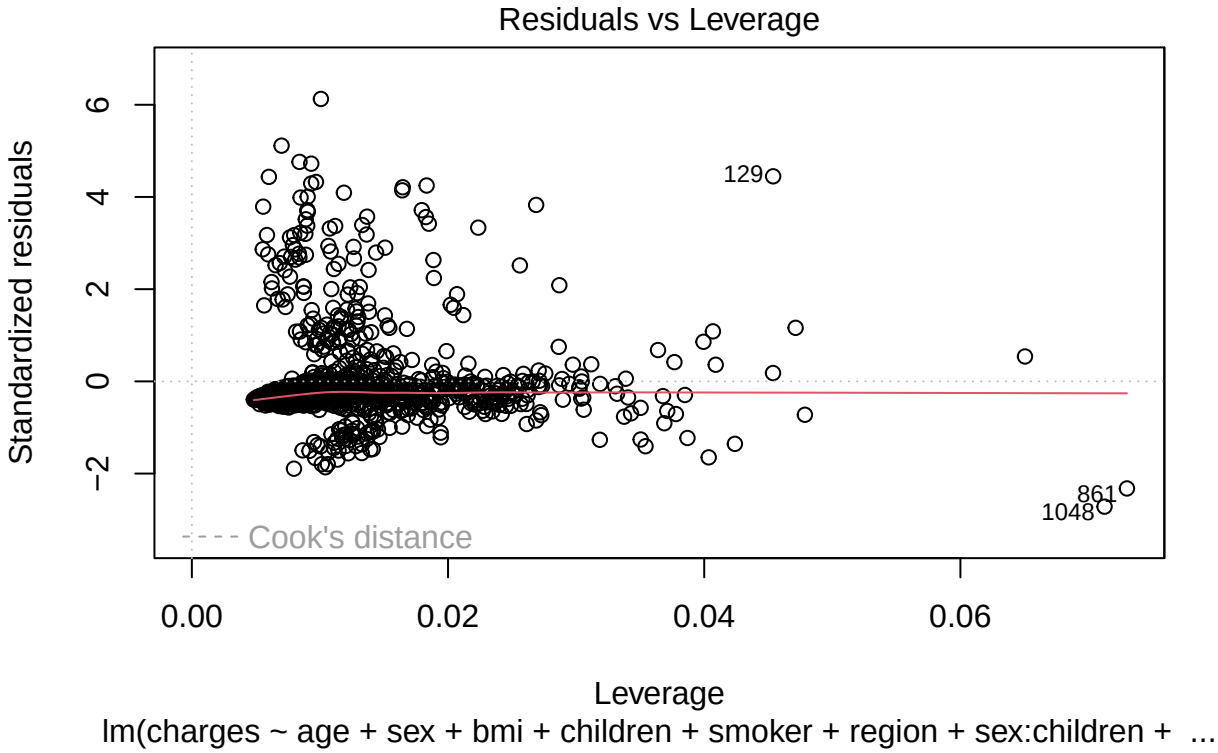
```
##
## Call:
## lm(formula = charges ~ age + sex + bmi + children + smoker +
##     region + sex:children + bmi:smoker + bmi:region, data = train_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12823.6  -1935.6  -1271.0   -302.6  29850.8
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)          -3184.93    1659.50  -1.919 0.055227 .
## age                    255.07      10.92  23.362  < 2e-16 ***
## sexmale                -70.60     408.71  -0.173 0.862892
## bmi                     61.11      54.05   1.131 0.258517
## children               704.68     181.78   3.877 0.000112 ***
## smokeryes           -20446.41    1846.34 -11.074  < 2e-16 ***
## regionnorthwest      -2033.77    2343.09  -0.868 0.385599
## regionsoutheast       3816.41    2165.83   1.762 0.078342 .
## regionsouthwest       -370.90    2317.12  -0.160 0.872858
## sexmale:children      -371.87     250.48  -1.485 0.137945
## bmi:smokeryes         1446.77      58.64  24.671  < 2e-16 ***
## bmi:regionnorthwest     42.60      78.66   0.542 0.588225
## bmi:regionsoutheast   -152.49      68.17  -2.237 0.025508 *
## bmi:regionsouthwest    -32.66      75.36  -0.433 0.664789
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4898 on 1056 degrees of freedom
## Multiple R-squared:  0.844,  Adjusted R-squared:  0.8421
## F-statistic: 439.5 on 13 and 1056 DF,  p-value: < 2.2e-16
```

```r
plot(aic_model)
```


Residuals vs Fitted
Residuals
Fitted values
lm(charges ~ age + sex + bmi + children + smoker + region + sex:children + ...
1301
517
243
30000
10000
0
−10000
0
10000
20000
30000
40000
50000
60000


Q−Q Residuals
Standardized residuals
Theoretical Quantiles
lm(charges ~ age + sex + bmi + children + smoker + region + sex:children + ...
1301
517
240
−2
0
2
4
6
−3
−2
−1
0
1
2
3


Scale−Location
1301
517
243
√|Standardized residuals|
0.0
0.5
1.0
1.5
2.0
2.5
0
10000
20000
30000
40000
50000
60000
Fitted values
lm(charges ~ age + sex + bmi + children + smoker + region + sex:children + ...


Residuals vs Leverage
Standardized residuals
6
4
2
0
−2
129
861
1048
Cook's distance
0.00
0.02
0.04
0.06
Leverage
lm(charges ~ age + sex + bmi + children + smoker + region + sex:children + ...